gEGNO: A Gated Equivariant Graph Neural Operator for Airflow Prediction

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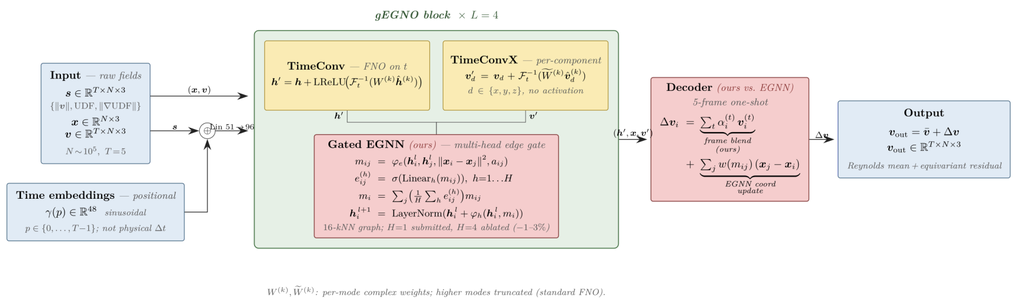

Fig. 1. **gEGNO architecture** — one block of the $L{=}4$ stack; red = our contributions. Time embeddings are positional indices, not physical Δt.

1 Abstract

- **Task.** Predict five future velocity frames on a 100k-point cloud surrounding one to three airfoils.
- **Model.** gEGNO couples an $E(3)$-equivariant EGNN backbone [3] with FNO-style spectral mixing along time [4] and a Reynolds mean-residual decoder [1].
- **Result.** Test L2 = **1.118**, reducing the organiser MLP baseline (3.16) by 65% and a persist-last-frame baseline (1.68) by 33%. Ablations at df = 0.15 show spectral temporal mixing as the largest single contribution (-5.3% test L2), with per-edge gating inside the full gEGNO stack adding a further -0.7%.

2 Problem & Symmetries

Given five consecutive frames of the 3D velocity field, predict the next five. Evaluation is mean per-point L2, averaged across the five output frames.

Symmetry	Consequence
$E(3)$-equivariance (Navier–Stokes)	EGNN scalar/vector separation
No-slip $v = 0$ on airfoil	Hard zero-masking at airfoil-surface nodes
Reynolds decomposition [1] $v = \bar{v} + v'$	Predict fluctuation Δv about the temporal mean $\bar{v}$

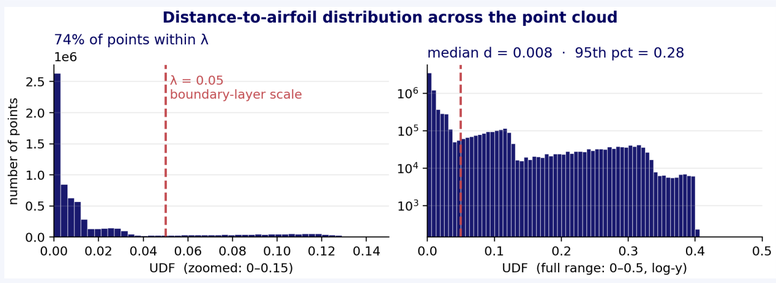

Fig. 2. Distance-to-airfoil distribution over 80 samples (8M points). **Left:** linear-y zoom of the boundary layer. 74% of points lie within $\lambda{=}0.05$, with median $d \approx 0.008$. **Right:** log-y across the full range, showing the free-stream tail.

3 Features & Graph

Let $s^\star(x_i) = \arg\min_{s\in\mathcal{S}} \|x_i - s\|_2$ denote the nearest airfoil-surface point.

Truncated UDF

$$\tilde{d}(x_i) = \min(\|x_i - s^\star\|_2,\ d_{\max}), \quad d_{\max} = 0.5$$

Inward surface normal (equivariant, $-\nabla\tilde{d}$)

$$\hat{n}(x_i) = \frac{s^\star - x_i}{\|s^\star - x_i\|_2}$$

Fixed kNN (submitted)

Directed k-NN with $k = 16$ in $\mathbb{R}^3$, computed per sample via SciPy's `cKDTree`. The graph topology is invariant under rigid motions.

UDF-adaptive kNN (ablated)

$$k(x_i) = k_{\text{far}} + (k_{\text{near}} - k_{\text{far}})\, e^{-\tilde{d}(x_i)/\lambda},\ \lambda{=}0.05$$

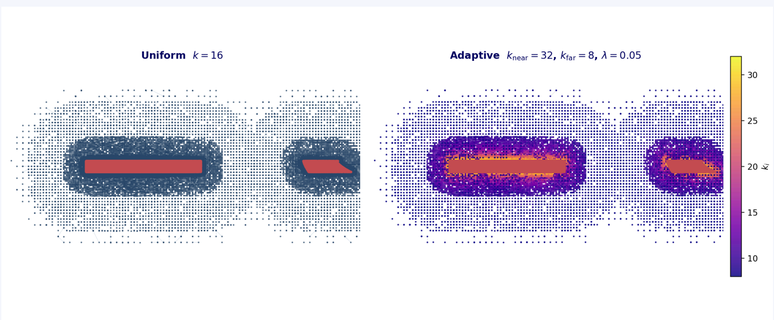

Fig. 3. Uniform $k{=}16$ (left) vs UDF-adaptive $k_{\text{near}}{=}32,\ k_{\text{far}}{=}8$ (right). The adaptive graph concentrates capacity near the boundary layer but comes out 1.6% worse on test with the plain EGNN backbone at df 0.15 (unswept).

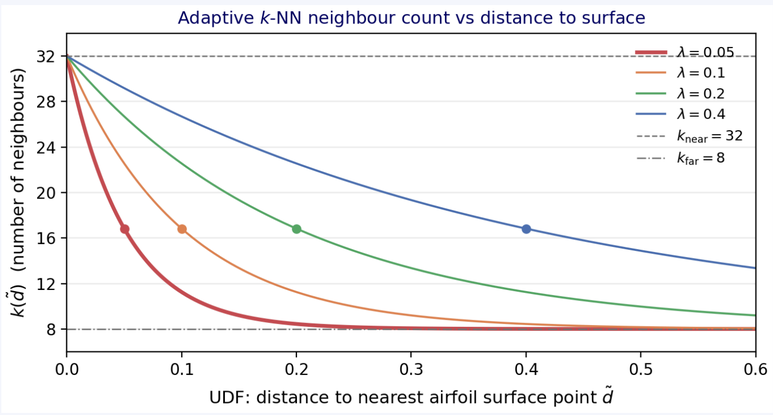

Fig. 4. Decay of $k(d)$ for several λ. $\lambda{=}0.05$ anchors the 1/e point at the empirical boundary-layer scale and saturates at $k_{\text{far}}{=}8$ by $d \approx 0.2$.

4 Architecture

Time-axis semantics (ours vs EGNO [5])

Ours. The time axis spans five real input frames. Positional indices $0\ldots4$ serve only as order tags, and the spectral operator mixes genuine temporal history.

EGNO (contrast). A single current state is replicated P times, each copy tagged with its physical future offset Δt_p. The spectral operator therefore mixes copies of the same state.

TimeConv (spectral mixing along t)

$$h' = h + \mathrm{LReLU}\big(\mathcal{F}_t^{-1}(W^{(k)}\hat{h}(k))\big)$$

TimeConvX (spectral mixing along t, applied independently to each velocity component)

$$v_d'(x_i,t) = v_d(x_i,t) + \mathcal{F}_t^{-1}(\tilde{W}^{(k)}\hat{v}_d(x_i,k))$$

EGNN with multi-head edge gate

$$m_{ij} = \phi_e(h_i^l, h_j^l, \|x_i - x_j\|^2, a_{ij})$$
$$e_{ij}^{(h)} = \sigma(\mathrm{Linear}_h(m_{ij})),\ h = 1,\ldots,H$$
$$m_i = \textstyle\sum_j \big(\frac{1}{H}\sum_h e_{ij}^{(h)}\big)\, m_{ij}$$
$$h_i^{l+1} = \mathrm{LayerNorm}(h_i^l + \phi_h(h_i^l, m_i))$$

Edge features a_{ij} (11 scalars): $\|x_i - x_j\|^2$ plus per-timestep projections $\langle v_i^{(t)}, \hat{r}_{ij}\rangle$ and $\langle v_j^{(t)}, \hat{r}_{ij}\rangle$ onto the invariant edge direction $\hat{r}_{ij}$. The red term is our multi-head edge-gate extension of EGNN [3, Eq. 7–8]. The submitted model uses $H = 1$; widening to $H = 4$ regressed test L2 by 1–3%.

Decoder (ours vs EGNN)

$$\Delta v_i = \underbrace{\textstyle\sum_t \alpha_i^{(t)} v_i^{(t)}}_{\text{learned frame blend}} + \underbrace{\textstyle\sum_j w(m_{ij})(x_j - x_i)}_{\text{EGNN coord update}}$$

The first (red) term is our learned per-frame blend. The second follows EGNN's coordinate update [3, Eq. 4]. All five output frames are decoded in a single pass, without autoregressive compounding.

5 Target & Loss

Following Reynolds [1], we predict the fluctuation Δv against the temporal mean $\bar{v}$ of the input window.

$$v_{\text{out}}(x_i,t) = \underbrace{\tfrac{1}{T}\textstyle\sum_{t'} v_{\text{in}}(x_i,t')}_{\bar{v}} + \Delta v(x_i,t)$$

Per-point MSE

$$\mathcal{L} = \tfrac{1}{T}\textstyle\sum_{t=0}^{T-1}\tfrac{1}{N}\sum_{i=1}^{N}\|\hat{v}_{\text{out}}(x_i,t) - v_{\text{out}}(x_i,t)\|_2^2$$

6 Training Setup

Optimizer	AdamW, lr = $2{\cdot}10^{-3}$, wd = 10^{-2}
Schedule	Linear warmup (15 ep) + cosine
Batch	1 × 8 grad-accum (eff. 8)
Epochs	120, patience 30 (val L2)
Split (final)	95 / 2.5 / 2.5%, geometry-stratified
Split (ablations)	70 / 15 / 15% within df 0.15, stratified
Time emb.	Sinusoidal [2]
Hardware (final)	1× NVIDIA B200 (180 GB), Modal, ~15 h 37 m
Hardware (ablations)	1× NVIDIA L40S, 30 epochs each

7 Results — Ablations

Each panel isolates one design choice at df = 0.15, where df is the fraction of the 770-file training split used.

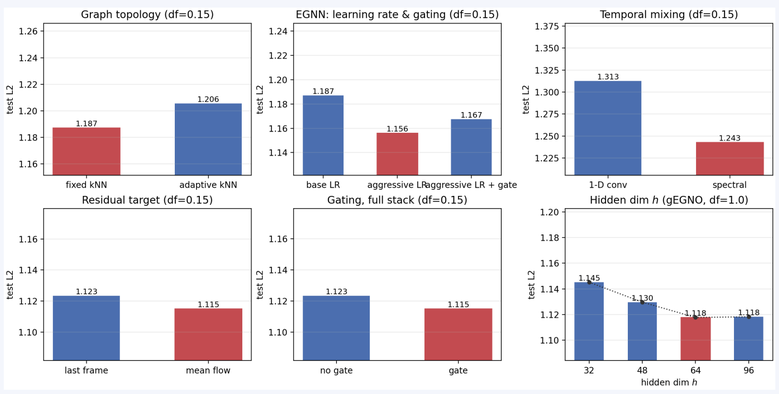

Fig. 5. Test L2 per ablation panel. The red bar marks the best variant within each panel.

- **Graph topology** (df = 0.15). Fixed kNN outperforms UDF-adaptive kNN by 1.6% on test L2.
- **Learning rate** (df = 0.15). The more aggressive learning rate improves test L2 by **2.6%**.
- **Temporal mixing** (df = 0.15). Spectral convolution outperforms 1-D time-conv by **5.3%**.
- **Residual target** (df = 0.15). Predicting the Reynolds mean-residual outperforms the last-frame residual by **0.7%**.
- **Gating, in full gEGNO** (df = 0.15). The per-edge gate outperforms the ungated variant by **0.73%** (1.1151 vs 1.1233).
- **Hidden dim** (df = 1.0). Test L2 plateaus at $h = 64$ (1.118). Widening to $h = 96$ lowers val L2 to 0.905 without changing test L2.

8 Qualitative Analysis of Errors

Spatial localisation across the rollout

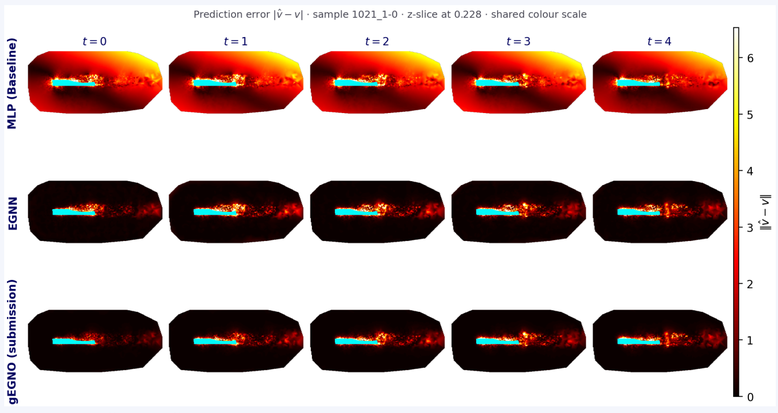

Fig. 6. Error $\|\hat{v} - v\|_2$ on sample 1021_1-0, shared colour scale. MLP error saturates everywhere. EGNN clears the free stream but leaves scattered wake noise. gEGNO further localises the residual into a narrow band behind the airfoil. Error grows by roughly $1.4\times$ across the five frames, consistent with one-shot decoding.

Per-frame error growth across time

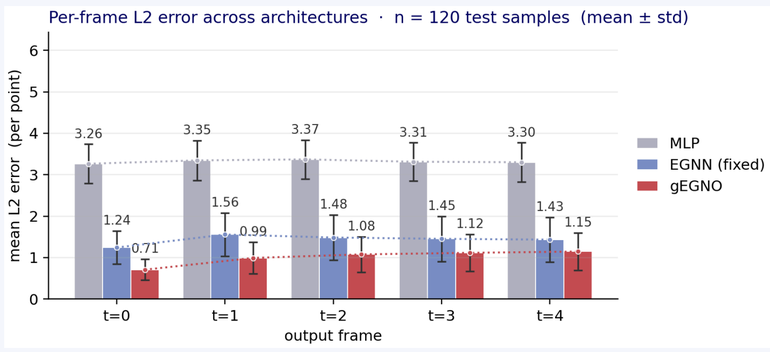

Fig. 7. Per-frame L2 (mean ± std across **n = 120** test samples). gEGNO has the lowest error at every timestep. At $t = 4$ it reaches 1.15, against 1.43 for EGNN (**−20%**) and 3.30 for MLP (**−65%**). Error growth is sub-exponential, consistent with one-shot decoding without autoregressive compounding. Caveat: the test split overlaps with the submission's training geometries, so gEGNO's reported margin is likely optimistic.

9 Limitations & Outlook

- **Gating vs attention.** The per-edge sigmoid is an unnormalised gate. Unlike GAT's softmax over $\mathcal{N}(i)$ [14], it allows selective muting but has no built-in self-regularisation. Inside the full gEGNO stack, gating reduces test L2 by **0.73%**. A direct comparison to SE(3)-Transformer [10] and EquiFormer [11] would test whether softmax normalisation outperforms our unnormalised gate.
- **Methodology caveats.** All ablations use a single seed (no confidence intervals) and the 70/15/15 split at df = 0.15. The submission itself was trained at df = 1.0 with a 95/2.5/2.5 split. Effective batch size is 8 via gradient accumulation.
- **Implementation note.** The submitted weights ignore the direction of `udf_gradient` (a bug). Only its magnitude reaches the encoder, and that magnitude is identically 1.
- **Under-explored directions.** We fixed depth at $d = 4$ without a principled sweep. The adaptive-kNN scale λ could not be swept within GPU memory limits. We did not test equivariant input noise as a symmetry-preserving augmentation.
- **Absolute-coordinate encodings.** Only UDF scalars and edge-direction vectors reach the encoder today. Fourier or rotary features on (x, y, z) would expose global position and may help resolve finer spatial structure.

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Acknowledgements

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GitHub Modal

Submitted configuration: gEGNO, h = 96, d = 4 blocks on fixed 16-kNN with the Reynolds mean-residual decoder. Spectral time-mixing with 3 Fourier modes (full length-5 real spectrum).